


Lecture # 9

Project Scheduling Management
Or
Project Time Management

ENGINEERING
PROJECT MANAGEMENT

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Project Scheduling Management

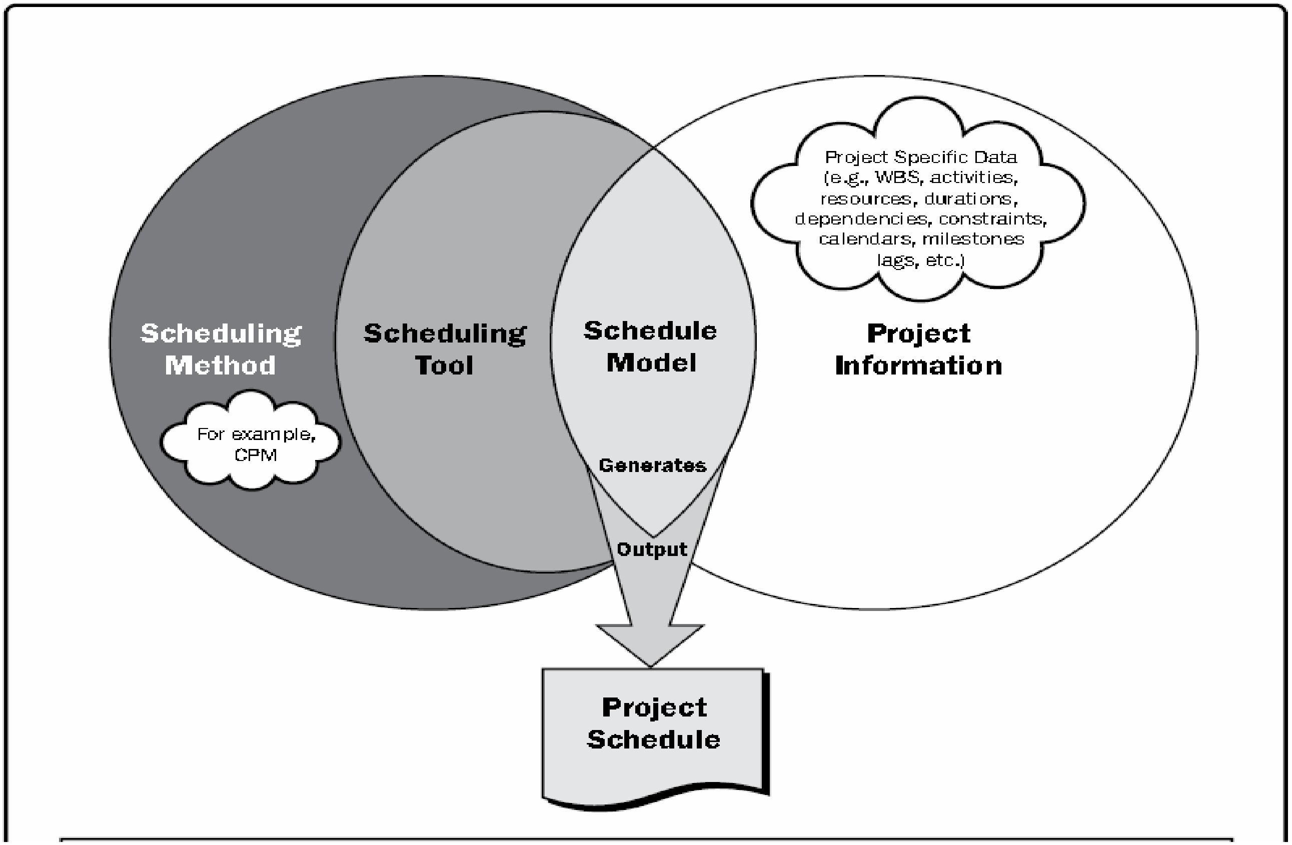
- 6.1 Plan Schedule Management—The process of establishing the policies, procedures, and documentation for planning, developing, managing, executing, and controlling the project schedule.
- 6.2 Define Activities—The process of identifying and documenting the specific actions to be performed on produce the project deliverables.
- 6.3 Sequence Activities—The process of identifying and documenting relationships among the project activities.
- 6.4 Estimate Activity Durations—The process of estimating the number of work period seeded to complete individual activities with the estimated resources.
- 6.5 Develop Schedule—The process of analyzing activity sequences, durations, resource requirements, and schedule constraints to create the project schedule model for project execution and monitoring and controlling.
- 6.6 Control Schedule—The process of monitoring the status of the project to update the project schedule and manage changes to the schedule baseline

Project Schedule Management

Table 1-4. Project Management Process Group and Knowledge Area Mapping

Knowledge Areas	Project Management Process Groups				
	Initiating Process Group	Planning Process Group	Executing Process Group	Monitoring and Controlling Process Group	Closing Process Group
4. Project Integration Management	4.1 Develop Project Charter	4.2 Develop Project Management Plan	4.3 Direct and Manage Project Work 4.4 Manage Project Knowledge	4.5 Monitor and Control Project Work 4.6 Perform Integrated Change Control	4.7 Close Project or Phase
5. Project Scope Management		5.1 Plan Scope Management 5.2 Collect Requirements 5.3 Define Scope 5.4 Create WBS		5.5 Validate Scope 5.6 Control Scope	
6. Project Schedule Management		6.1 Plan Schedule Management 6.2 Define Activities 6.3 Sequence Activities 6.4 Estimate Activity Durations 6.5 Develop Schedule		6.6 Control Schedule	
7. Project Cost Management		7.1 Plan Cost Management 7.2 Estimate Costs 7.3 Determine Budget		7.4 Control Costs	
8. Project Quality Management		8.1 Plan Quality Management	8.2 Manage Quality	8.3 Control Quality	

Project Schedule Management



PLAN SCHEDULE MANAGEMENT

- The key benefit of this process is that it provides guidance and direction on how the project schedule will be managed throughout the project

PLAN SCHEDULE MANAGEMENT

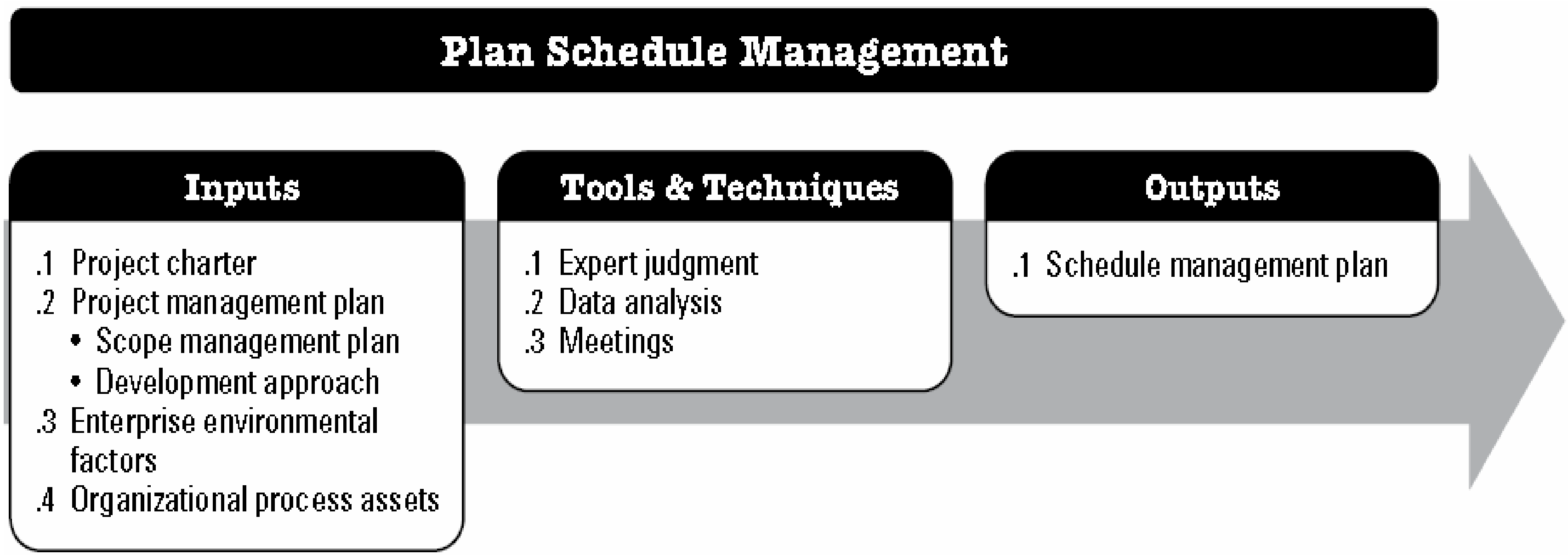
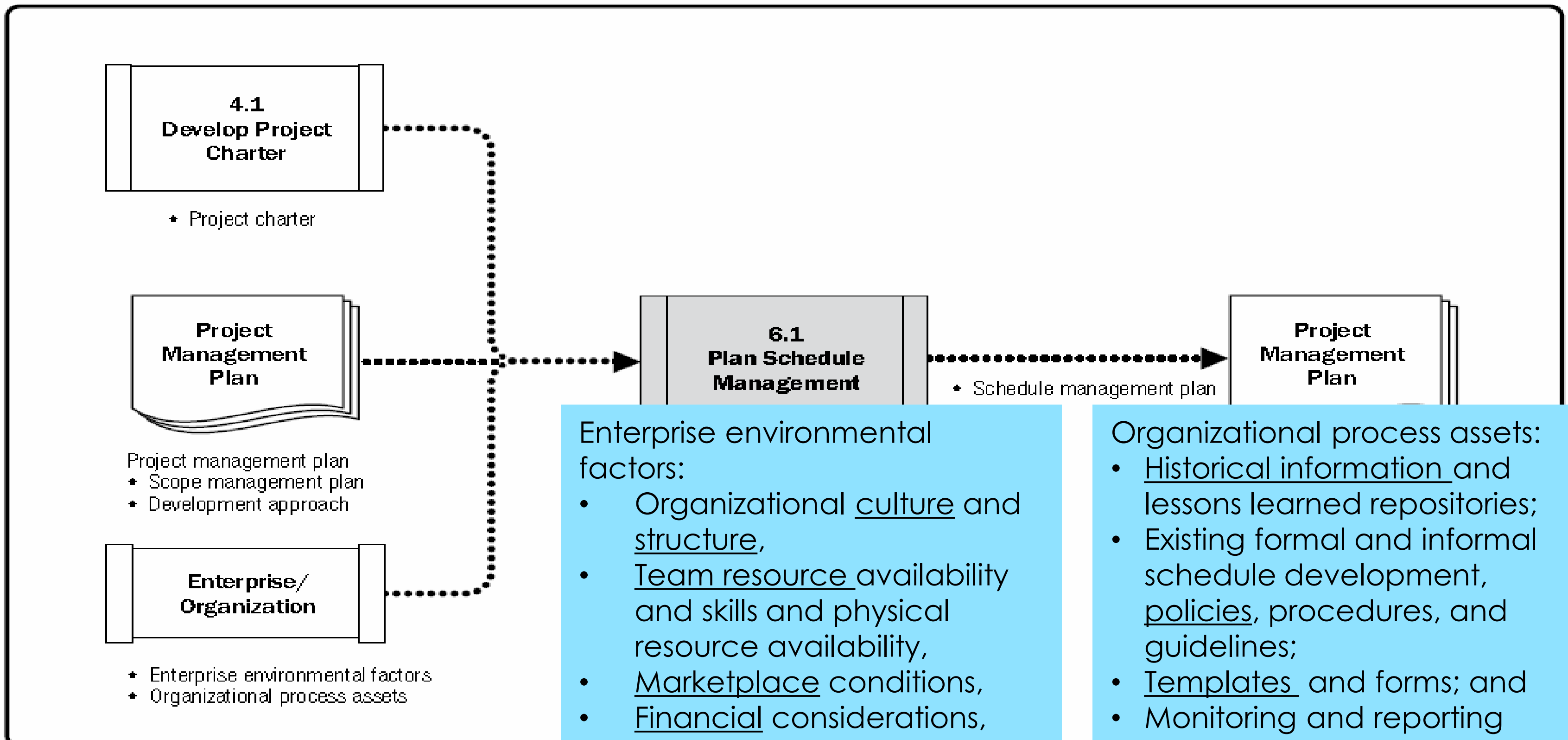


Figure 6-3. Plan Schedule Management: Inputs, Tools & Techniques, and Outputs

PLAN SCHEDULE MANAGEMENT



Enterprise environmental factors:

- Organizational culture and structure,
- Team resource availability and skills and physical resource availability,
- Marketplace conditions,
- Financial considerations,
- Social influences

Organizational process assets:

- Historical information and lessons learned repositories;
- Existing formal and informal schedule development, policies, procedures, and guidelines;
- Templates and forms; and
- Monitoring and reporting tools

Figure 6-4. Plan Schedule Management: Data Flow Diagram

Define Activities

- The key benefit of this process is that it provides guidance and direction on how scope will be managed throughout the The key benefit of this process is that it decomposes work packages into schedule activities that provide a basis for estimating scheduling, executing, monitoring, and controlling the project work project

Define Activities

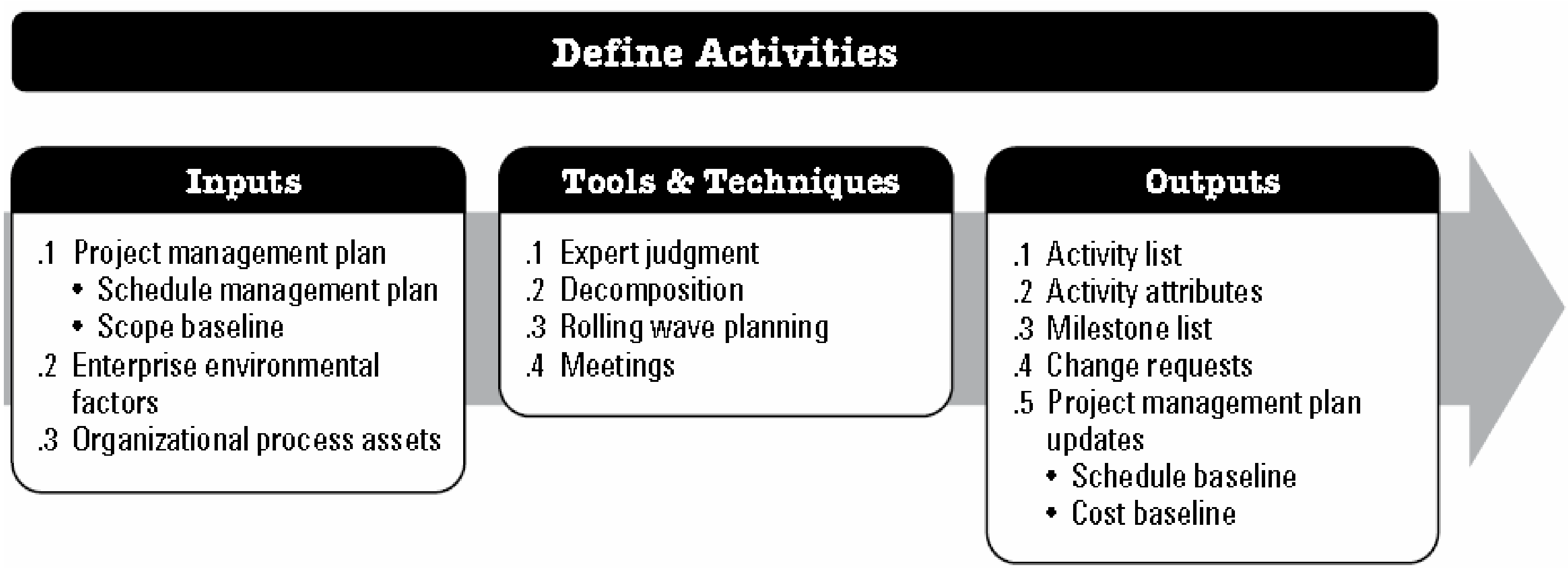


Figure 6-5. Define Activities: Inputs, Tools & Techniques, and Outputs

SEQUENCE ACTIVITIES

- The key benefit of this process is that it defines the logical sequence of work to obtain the greatest efficiency given all project constraints.

Sequence Activities

Sequence Activities

Inputs

.1 Project management plan
- Schedule management plan
- Scope baseline

.2 Project documents
- Activity attributes
- Activity list
- Assumption log
- Milestone list

.3 Enterprise environmental factors

.4 Organizational process assets

Tools & Techniques

.1 Precedence diagramming method

.2 Dependency determination and integration

.3 Leads and lags

.4 Project management information system

Outputs

.1 Project schedule network diagrams

.2 Project documents updates
- Activity attributes
- Activity list
- Assumption log
- Milestone list

Figure 6-7. Sequence Activities: Inputs, Tools & Techniques, and Outputs

Precedence Diagramming Method - PDM

- The precedence diagramming method (PDM) is
  - A technique used for constructing a schedule model
  - Activities are represented by nodes
  - Are graphically linked by one or more logical relationships to show the sequence in which the activities are to be performed.
  - A predecessor activity is an activity that logically comes before a dependent activity in a schedule.
  - A successor activity is a dependent activity that logically comes after another activity in a schedule.

Precedence Diagramming Method - PDM

- The PDM includes four types of dependencies or logical relationships. They are:
  - **Earliest Start Time or Early Start (ES).** This is the earliest time an activity can be started in your project.
  - **Latest Start Time or Late Start Time (LS).** This is the latest time that an activity can be started on your project. If you start the activity beyond this time, it will affect your critical path.,
  - **Earliest Finish Time or Early Finish (EF).** This is the earliest time an activity is completed in your project.
  - **Latest Finish Time of Late Finish (LF)**This is the latest time you can complete the activity on your project. If your activity crosses this time, your project will be delayed.

ESTIMATE ACTIVITY DURATIONS

- The key benefit of this process is that it provides the amount of time each activity will take to complete

ESTIMATE ACTIVITY DURATIONS

Estimate Activity Durations

Inputs

.1 Project management plan
- Schedule management plan
- Scope baseline

.2 Project documents
- Activity attributes
- Activity list
- Assumption log
- Lessons learned register
- Milestone list
- Project team assignments
- Resource breakdown structure
- Resource calendars
- Resource requirements
- Risk register

.3 Enterprise environmental factors

.4 Organizational process assets

Tools & Techniques

.1 Expert judgment

.2 Analogous estimating

.3 Parametric estimating

.4 Three-point estimating

.5 Bottom-up estimating

.6 Data analysis
- Alternatives analysis
- Reserve analysis

.7 Decision making

.8 Meetings

Outputs

.1 Duration estimates

.2 Basis of estimates

.3 Project documents updates
- Activity attributes
- Assumption log
- Lessons learned register

Figure 6-12. Estimate Activity Durations: Inputs, Tools & Techniques, and Outputs

ESTIMATE ACTIVITY DURATIONS

- **Number of resources**
  - Increasing the number of resources to twice the original number of the resources does not always reduce the time by half, as it **may increase extra duration** due to
  - Risk, knowledge transfer, learning curve, additional coordination, and other factors involved
- **Advances in technology**
  - For example, an increase in the output of a manufacturing plant may be achieved by procuring the latest advances in technology, which may impact duration and resource needs
- **Motivation of staff.**
  - The project manager also needs to be aware of
  - Student Syndrome—or **procrastination**— when people start to apply themselves only at the last possible moment before the deadline, and
  - Parkinson's Law where work expands to fill the time available for its completion

Develop Schedule

- The key benefit of this process is that it generates a schedule model with planned dates for completing project activities.

Develop Schedule

Develop Schedule

Inputs

.1 Project management plan
- Schedule management plan
- Scope baseline

.2 Project documents
- Activity attributes
- Activity list
- Assumption log
- Basis of estimates
- Duration estimates
- Lessons learned register
- Milestone list
- Project schedule network diagrams
- Project team assignments
- Resource calendars
- Resource requirements
- Risk register

.3 Agreements
.4 Enterprise environmental factors
.5 Organizational process assets

Tools & Techniques

.1 Schedule network analysis
.2 Critical path method
.3 Resource optimization
.4 Data analysis
- What-if scenario analysis
- Simulation

.5 Leads and lags
.6 Schedule compression
.7 Project management information system
.8 Agile release planning

Outputs

.1 Schedule baseline
.2 Project schedule
.3 Schedule data
.4 Project calendars
.5 Change requests
.6 Project management plan updates
- Schedule management plan
- Cost baseline

.7 Project documents updates
- Activity attributes
- Assumption log
- Duration estimates
- Lessons learned register
- Resource requirements
- Risk register

Figure 6-14. Develop Schedule: Inputs, Tools & Techniques, and Outputs

Tools and Techniques

- CRITICAL PATH METHOD (CPM) :
- The critical path method is used to
  - Estimate the minimum project duration and
  - Determine the amount of schedule flexibility on the logical network paths within the schedule model.
- This schedule network analysis technique calculates the:
  - Early start, early finish, late start, and late finish dates for all activities
  - Without regard for any resource limitations by performing a **forward** and **backward** pass analysis through the schedule network,
- The longest path on a network diagram … is the **critical path**
- The critical path is the sequence of activities that represents the longest path through a project, which determines the shortest possible project duration.
- The longest path has the **least total float—usually zero**.

Tools and Techniques

- The total float or schedule flexibility is measured by:
- the amount of time that a schedule activity can be delayed or extended from its early start date without delaying the project finish date or violating a schedule constraint.
- A critical path is normally characterized by zero total float on the critical path.
- the free float is the amount of time that
- a schedule activity can be delayed without delaying the early start date of any successor or violating a schedule constraint.
- For example the free float for Activity B, in Figure 6-16, is 5 days.

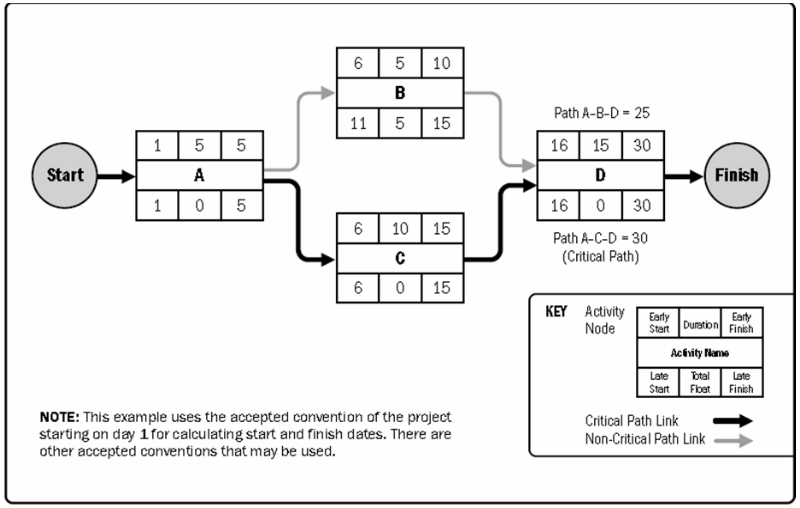


Figure 6-16. Example of Critical Path Method

Develop Schedule - CPM

Total Float = LS – ES or LF - EF

Free Float = $ES_{(successor)} - EF_{(current)} + 1$

The Free Float and the Total Float of a critical activity are = ZEROs

Tools and Techniques

(2) RESOURCE OPTIMIZATION :

- Examples of resource optimization techniques that can be used to adjust the schedule model :
  - **Resource leveling**. A technique in which start and finish dates are adjusted
    - based on resource constraints with the goal of balancing the demand for resources with the available supply is used when required resources are
    - available only at certain times or in limited quantities, or
    - over allocated, such as when a resource has been assigned to two or more activities during the same time period, or
    - OBJECTIVE: there is a need to keep resource usage at a constant level.
    - Resource leveling can often cause the original critical path to change. Available float is used for leveling resources. Consequently, the critical path through the project schedule may change.
  - **Resource smoothing**. A technique that adjusts the activities of a schedule model such that
    - OBJECTIVE: the requirements for resources on the project do not exceed certain predefined resource limits.
    - In resource smoothing, as opposed to resource leveling, the project's critical path is not changed and the completion date may not be delayed.

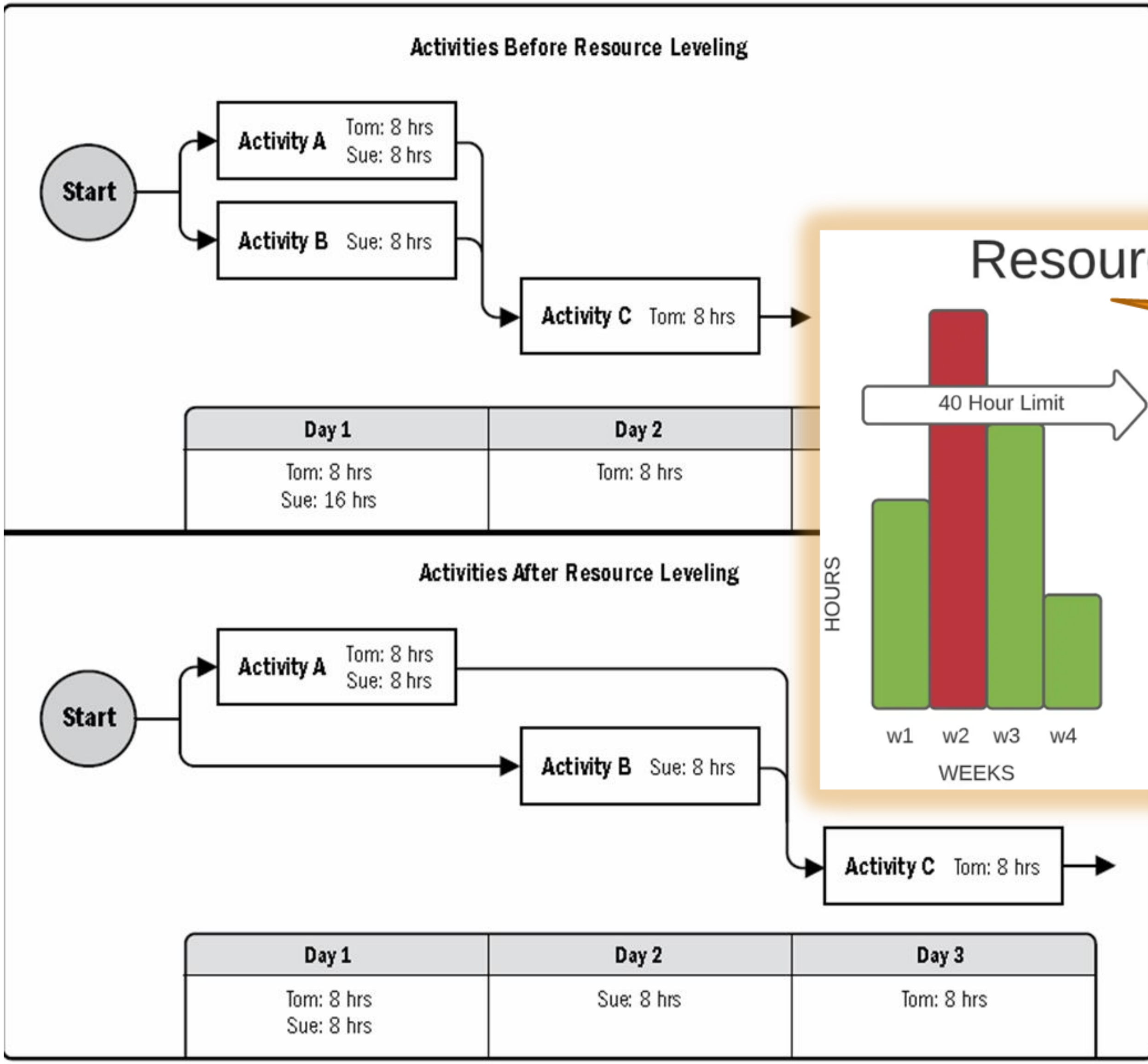


Figure 6-17. Resource Leveling

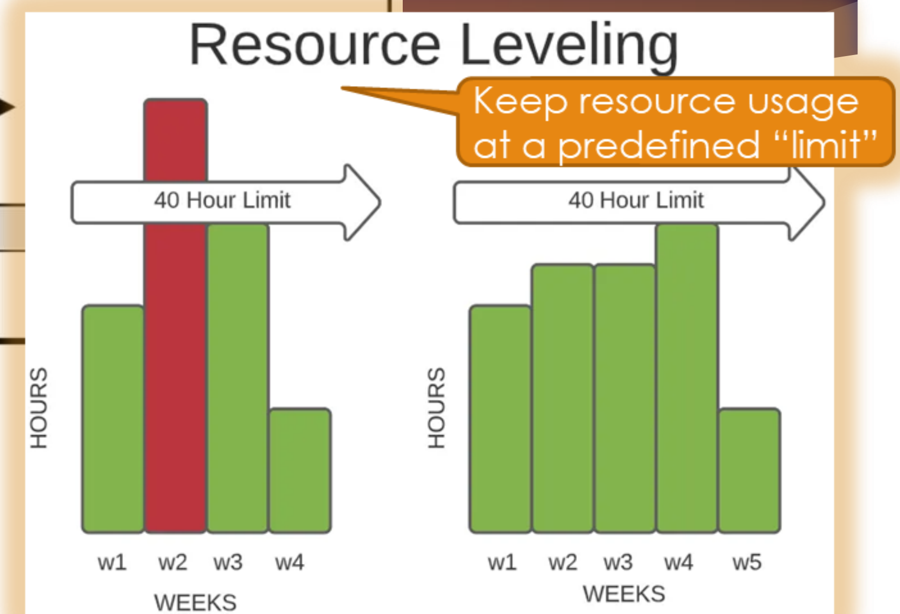
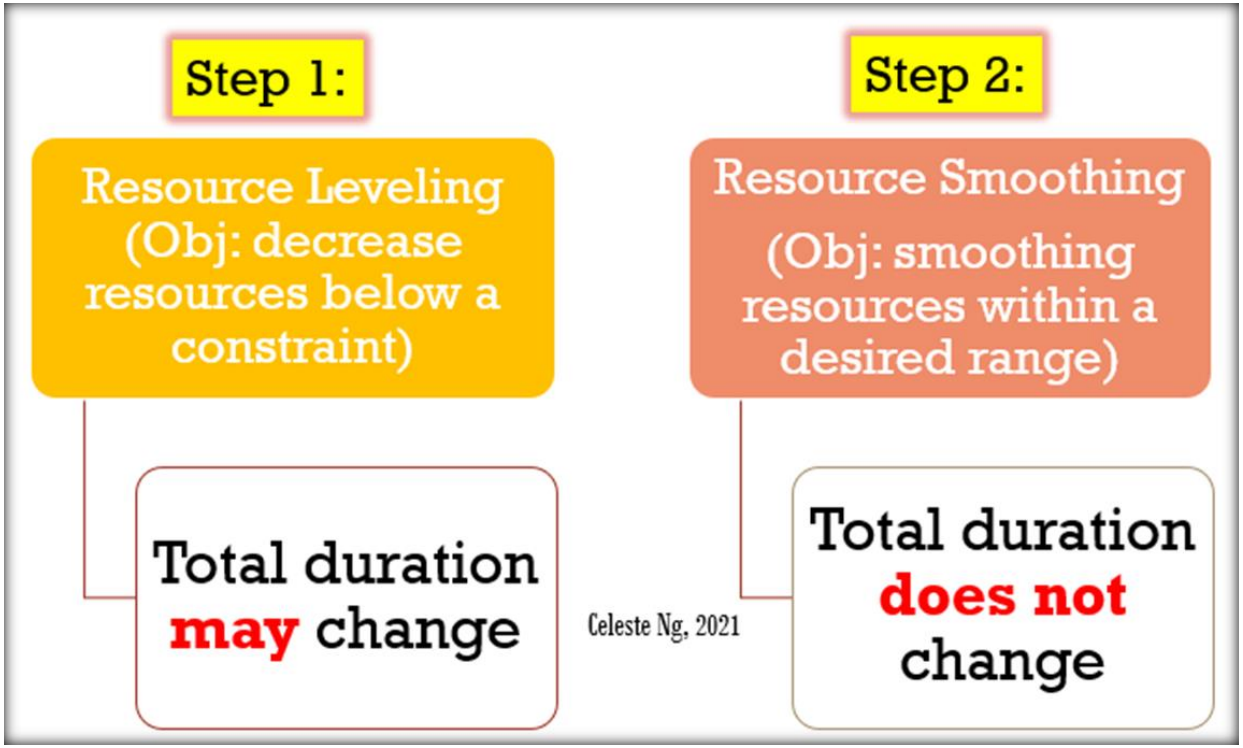


Photo credit: https://pm-training.net/resource-leveling-vs-resource-smoothing/

Develop Schedule – Resource Optimization



Resource Leveling

- Pros
  - Its ability to minimize deficits
  - Efficient use of resources
  - Prevents over-allocation
  - Ensures successful delivery of quality outputs
- Cons
  - Can cause delays in other projects
  - Budget over-runs
  - May demand extra resources

Resource Smoothing

- Pros
  - Successful and on-time project delivery
  - It is an efficient and cost-effective way to use resources
- Cons
  - Project managers will have to delay some work to give the particular project first priority.
  - To prevent delays, the schedule usually has little flexibility

Develop Schedule

(3) SCHEDULE COMPRESSION

- Schedule compression techniques are used to shorten or accelerate the schedule duration without reducing the project scope in order to meet schedule constraints, imposed dates, or other schedule objectives.
- Schedule compression techniques are:
  - **Crashing.** A technique used to shorten the schedule duration for the least incremental cost by adding resources.
  - Examples of crashing include approving overtime, bringing in additional resources, or paying to expedite delivery to activities on the critical path.
  - Crashing works only for activities on the critical path where additional resources will shorten the activity's duration.
  - Does not always produce a viable alternative and may result in increased risk and/or cost.
  - **Fast tracking**. A schedule compression technique in which activities or phases normally done in sequence
  - Fast tracking may result in rework, and increased risk and project costs.
  - Fast tracking only works when activities can be overlapped to shorten the project duration on the critical path.

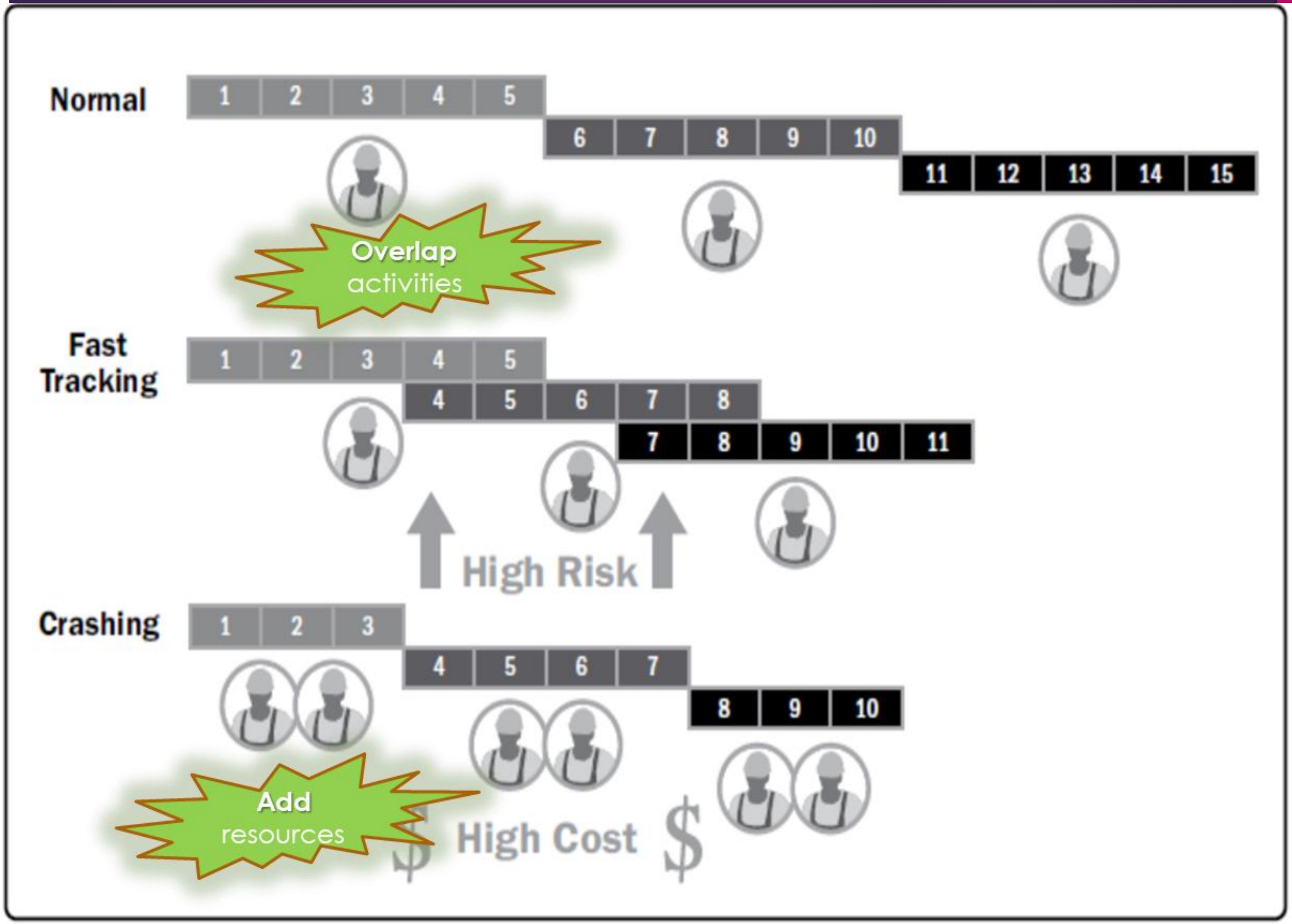
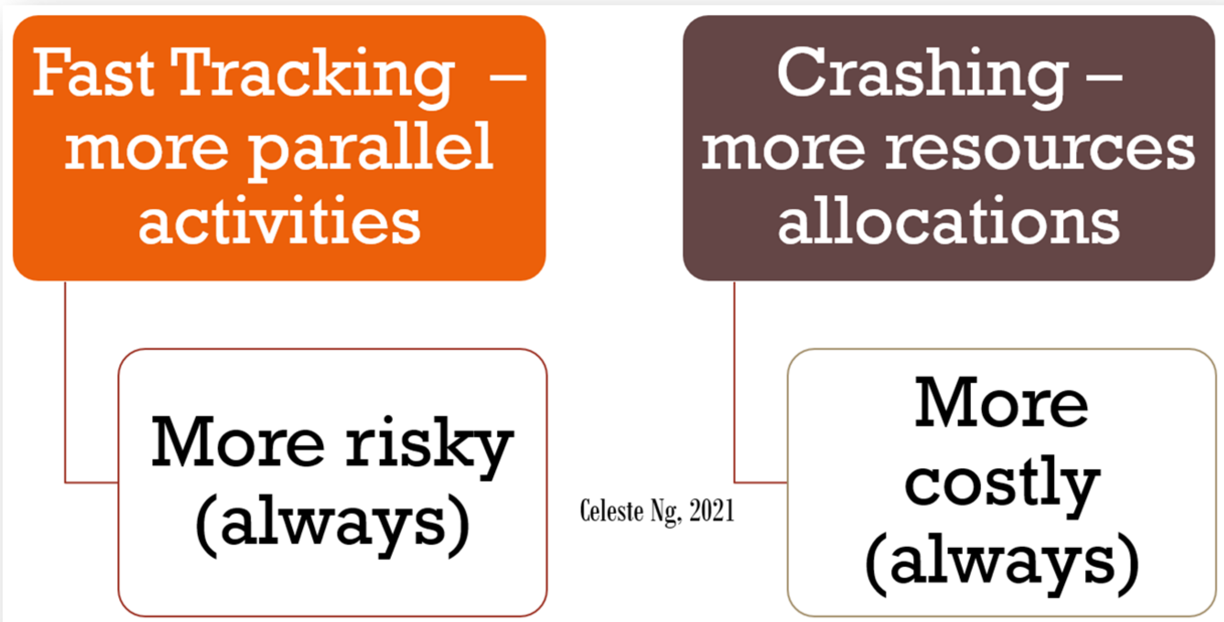


Figure 6-19. Schedule Compression Comparison

Develop Schedule – Schedule Compression



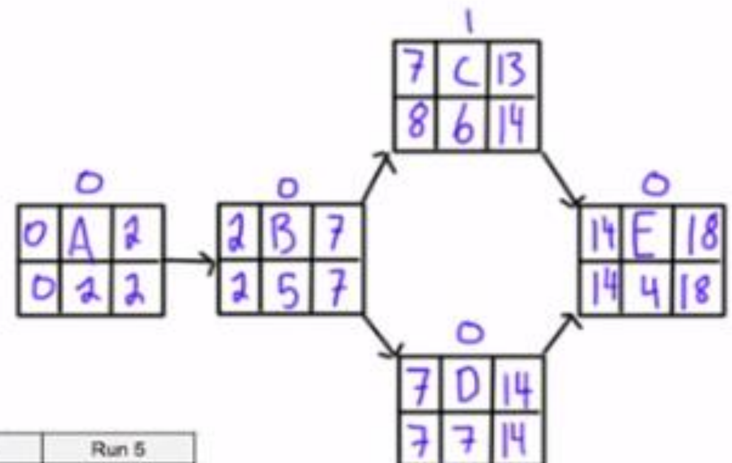
Develop Schedule - Crashing

Project Overhead Cost: $600/day → $600 \times 18 = 10,800$

Activity	Predecessor	Normal Duration	Crash Duration	Normal Cost of Activity ($)	Additional Cost/day to Crash ($)
A	-	2	2	1,200	0
B	A	5	4	1,800	600
C	B	6	4	1,700	400
D	B	7	5	1,500	700
E	E,F	4	3	1,900	500

8,100

$10,800 + 8100 = 18,900$



Activity	Initially				Run 1			Run 2			Run 3			Run 4			Run 5		
	Normal Duration	Crash Duration	Cost / Day	TF	Days Saved	Dur	TF	Days Saved	Dur	TF	Days Saved	Dur	TF	Days Saved	Dur	TF	Days Saved	Dur	TF
A	2	2	0	0															
B	5	4	600	0															
C	6	4	400	1															
D	7	5	700	0															
E	4	3	600	0															
Cost																			
Savings																			
Project Cost	18,900																		
Project Duration	18																		

Develop Schedule - Crashing

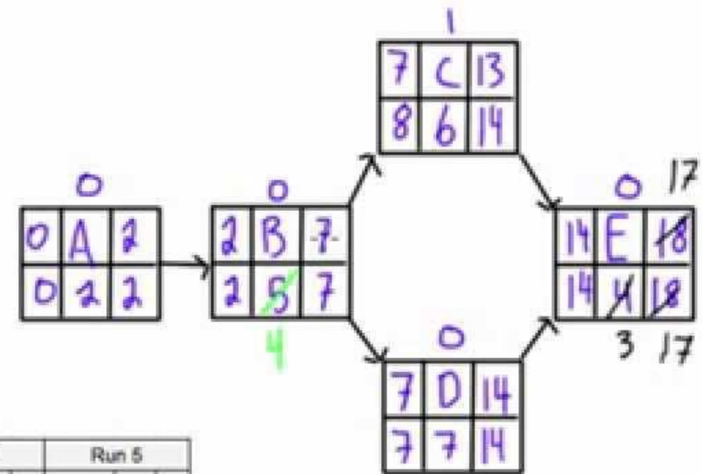
→ 600 × 18 = 10,800

Project Overhead Cost: $600/day

Activity	Predecessor	Normal Duration	Crash Duration	Normal Cost of Activity ($)	Additional Cost/day to Crash ($)
~~A~~	-	2	2	1,200	0
B	A	5	4	1,800	600
C	B	6	4	1,700	400
D	B	7	5	1,500	700
~~E~~	E,F	~~4~~ 3	3	1,900	500

8,100

10,800 + 8100 = 18,900



Activity	Initially				Run 1			Run 2			Run 3			Run 4			Run 5		
	Normal Duration	Crash Duration	Cost / Day	TF	Days Saved	Dur	TF	Days Saved	Dur	TF	Days Saved	Dur	TF	Days Saved	Dur	TF	Days Saved	Dur	TF
A	2	2	0	0		2	0		2										
B	5	4	600	0		5	0	1	4										
C	6	4	400	1		6	1		6										
D	7	5	700	0		7	0		7										
E	4	3	600	0	1	3	0		3										
Cost	—				500														
Savings	—				600														
Project Cost	18,900				18,800														
Project Duration	18				17														

Develop Schedule - Crashing

Project Overhead Cost: $600/day → 600 × 18 = 10,800

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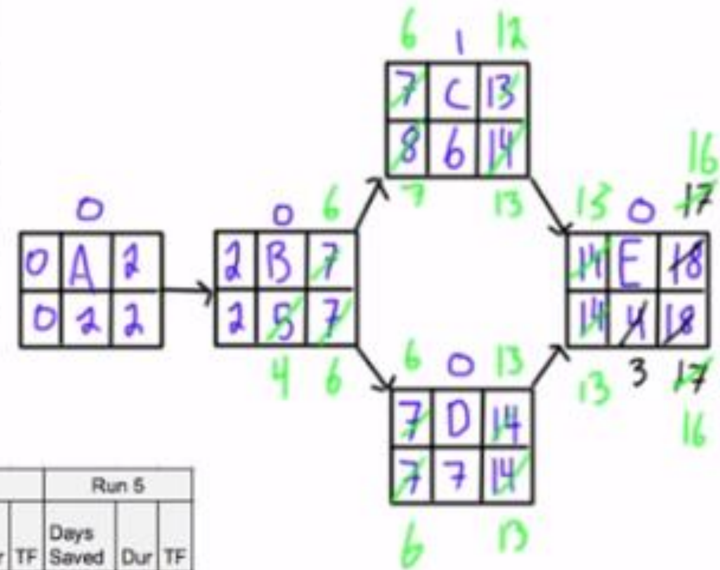

Activity	Initially				Run 1			Run 2			Run 3			Run 4			Run 5		
	Normal Duration	Crash Duration	Cost / Day	TF	Days Saved	Dur	TF	Days Saved	Dur	TF	Days Saved	Dur	TF	Days Saved	Dur	TF	Days Saved	Dur	TF
A	2	2	0	0		2	0		2	0									
B	5	4	600	0		5	0	1	4	1									
C	6	4	400	1		6	1		6	0									
D	7	5	700	0		7	0		7	0									
E	4	3	600	0	1	3	0		3	0									
Cost	—				500			600											
Savings	—				600			600											
Project Cost	18,900				18,800			18,800											
Project Duration	18				17			16											

Develop Schedule - Crashing

Project Overhead Cost: $600/day → 600 × 18 = 10,800

Activity	Predecessor	Normal Duration	Crash Duration	Normal Cost of Activity ($)	Additional Cost/day to Crash ($)
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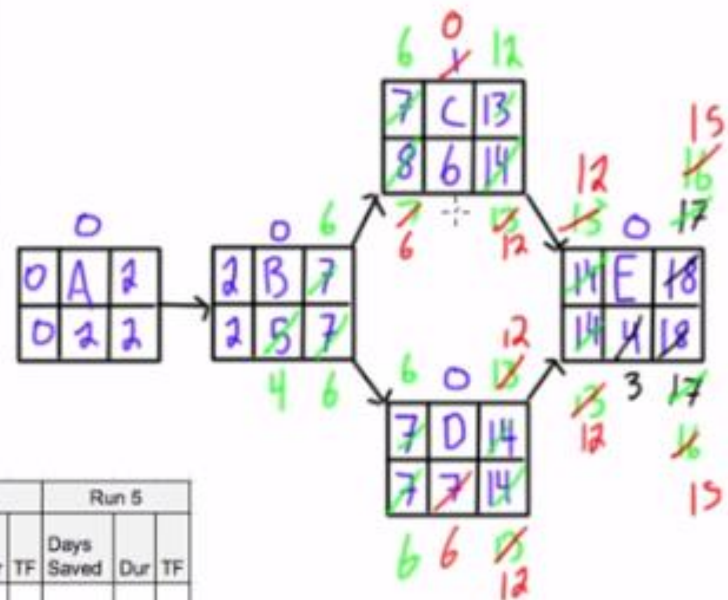

Activity	Initially				Run 1			Run 2			Run 3			Run 4			Run 5		
	Normal Duration	Crash Duration	Cost / Day	TF	Days Saved	Dur	TF	Days Saved	Dur	TF	Days Saved	Dur	TF	Days Saved	Dur	TF	Days Saved	Dur	TF
A	2	2	0	0		2	0		2	0		2	0						
B	5	4	600	0		5	0	1	4	0		4	0						
C	6	4	400	1		6	1		6	1		6	0						
D	7	5	700	0		7	0		7	0	1	6	0						
E	4	3	600	0	1	3	0		3	0		3	0						
Cost	—				500			600			700								
Savings	—				600			600			600								
Project Cost	18,900				18,800			18,800			18,900								
Project Duration	18				17			16			15								

Develop Schedule - Crashing

Project Overhead Cost: $600/day → 600 × 18 = 10,800

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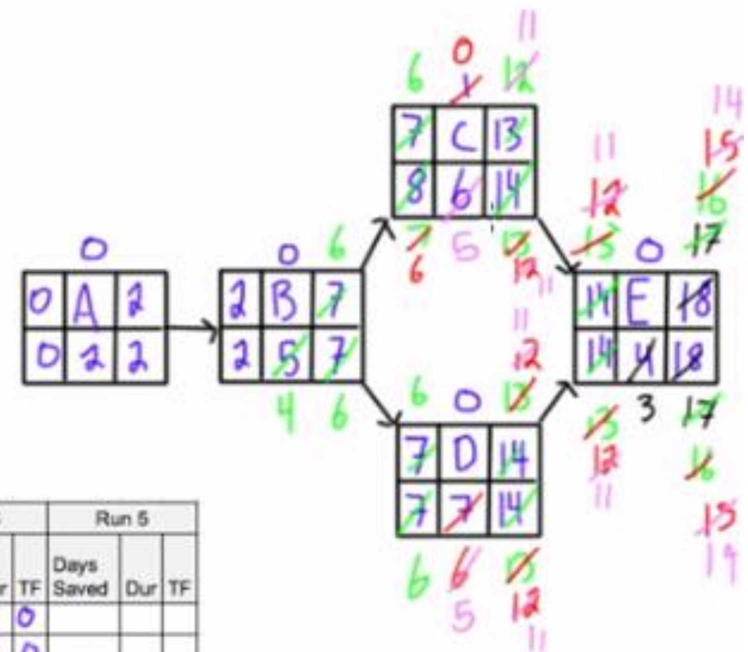

	Initially				Run 1			Run 2			Run 3			Run 4			Run 5		
Activity	Normal Duration	Crash Duration	Cost / Day	TF	Days Saved	Dur	TF	Days Saved	Dur	TF	Days Saved	Dur	TF	Days Saved	Dur	TF	Days Saved	Dur	TF
A	2	2	0	0		2	0		2	0		2	0		2	0			
B	5	4	600	0		5	0	1	4	0		4	0		4	0			
C	6	4	400	1		6	1		6	1		6	0	1	5	0			
D	7	5	700	0		7	0		7	0	1	6	0	1	5	0			
E	4	3	600	0	1	3	0		3	0		3	0		3	0			
Cost	—				500			600			700			11,800					
Savings	—				600			600			600			600					
Project Cost	18,900				18,800			18,800			18,900			19,420					
Project Duration	18				17			16			15			14					

Develop Schedule - Crashing

Project Overhead Cost: $600/day → 600 × 18 = 10, 800

Activity	Predecessor	Normal Duration	Crash Duration	Normal Cost of Activity ($)	Additional Cost/day to Crash ($)
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B	A	~~5~~ 4	4	1,800	600
C	B	~~6~~ 5	4	1,700	400
D	B	~~7~~ ~~6~~ 5	5	1,500	700
E	E,F	~~4~~ 3	3	1,900	500

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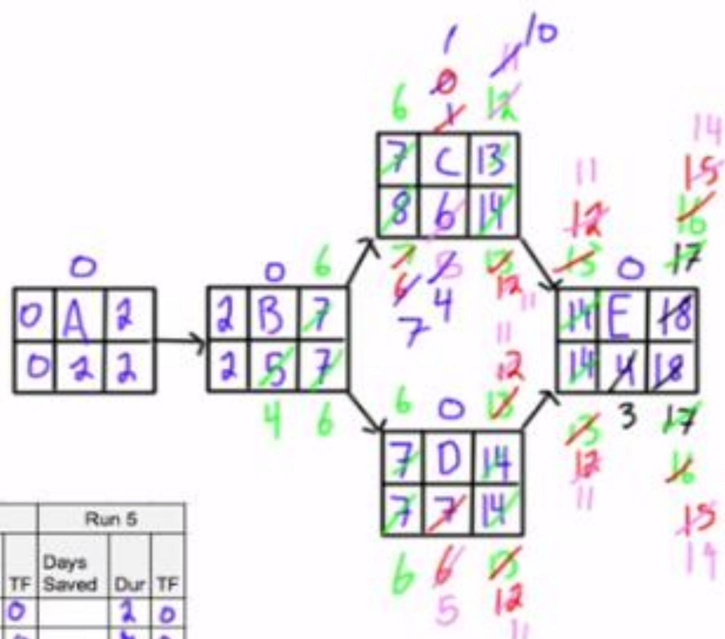
10,800 + 8100 = 18,900



Activity	Initially: Normal Duration	Initially: Crash Duration	Initially: Cost / Day	Initially: TF	Run 1: Days Saved	Run 1: Dur	Run 1: TF	Run 2: Days Saved	Run 2: Dur	Run 2: TF	Run 3: Days Saved	Run 3: Dur	Run 3: TF	Run 4: Days Saved	Run 4: Dur	Run 4: TF	Run 5: Days Saved	Run 5: Dur	Run 5: TF
A	2	2	0	0		2	0		2	0		2	0		2	0		2	0
B	5	4	600	0		5	0	1	4	0		4	0		4	0		4	0
C	6	4	400	1		6	1		6	1		6	0	1	5	0	1	4	1
D	7	5	700	0		7	0		7	0	1	6	0	1	5	0		5	0
E	4	3	600	0	1	3	0		3	0		3	0		3	0		3	0
Cost	—				500			600			700			11,800			400		
Savings	—				600			600			600			600			0		
Project Cost	18,900				18,800			18,800			18,900			19,420			19,800		
Project Duration	18				17			16			15			14			14		

Control Schedule

- The key benefit of this process is that the schedule baseline is maintained throughout the project.

Control Schedule

Control Schedule

Inputs

.1 Project management plan
- Schedule management plan
- Schedule baseline
- Scope baseline
- Performance measurement baseline

.2 Project documents
- Lessons learned register
- Project calendars
- Project schedule
- Resource calendars
- Schedule data

.3 Work performance data

.4 Organizational process assets

Tools & Techniques

.1 Data analysis
- Earned value analysis
- Iteration burndown chart
- Performance reviews
- Trend analysis
- Variance analysis
- What-if scenario analysis

.2 Critical path method

.3 Project management information system

.4 Resource optimization

.6 Leads and lags

.7 Schedule compression

Outputs

.1 Work performance information

.2 Schedule forecasts

.3 Change requests

.4 Project management plan updates
- Schedule management plan
- Schedule baseline
- Cost baseline
- Performance measurement baseline

.5 Project documents updates
- Assumption log
- Basis of estimates
- Lessons learned register
- Project schedule
- Resource calendars
- Risk register
- Schedule data

Figure 6-22. Control Schedule: Inputs, Tools & Techniques, and Outputs